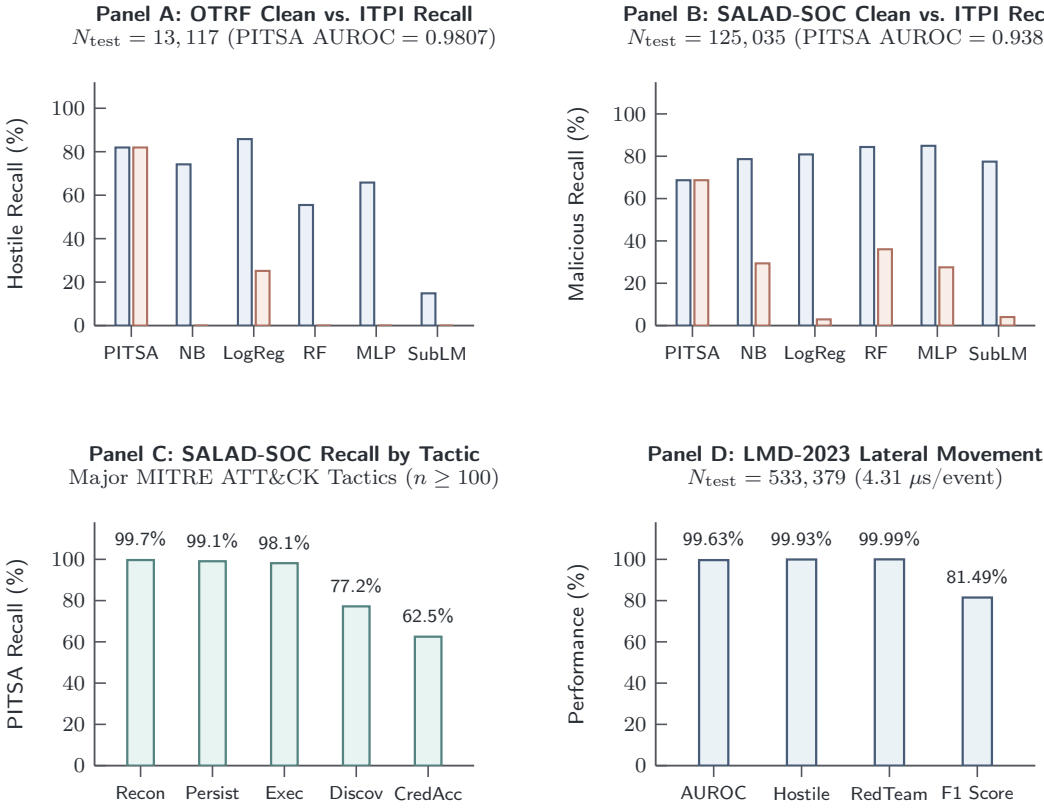

Panel A: OTRF Clean vs. ITPI Recall
$N_{\text{test}} = 13,117$ (PITSA AUROC = 0.9807)
Hostile Recall (%)
0
20
40
60
80
100
PITSA
NB
LogReg
RF
MLP
SubLM
Panel B: SALAD-SOC Clean vs. ITPI Recall
$N_{\text{test}} = 125,035$ (PITSA AUROC = 0.9380)
Malicious Recall (%)
0
20
40
60
80
100
PITSA
NB
LogReg
RF
MLP
SubLM
Panel C: SALAD-SOC Recall by Tactic
Major MITRE ATT&CK Tactics ($n \geq 100$)
PITSA Recall (%)
0
20
40
60
80
100
99.7%
99.1%
98.1%
77.2%
62.5%
Recon
Persist
Exec
Discov
CredAcc
Panel D: LMD-2023 Lateral Movement
$N_{\text{test}} = 533,379$ (4.31 μs/event)
Performance (%)
0
20
40
60
80
100
99.63%
99.93%
99.99%
81.49%
AUROC
Hostile
RedTeam
F1 Score
Panels A & B Bars: Blue = Clean Telemetry; Peach = ITPI-Perturbed Text ($\mathcal{A}_{\text{text}}$)
PITSA Invariance: 0.00 pp Recall Drop